Le-Space

SPIN-OFF B · CONCEPT

WebRTC Public Data Map

Where direct connections hold — and where a relay has to step in

> Every connection is a measurement. Millions of measurements tell you where you need a relay.

Demo Source npm

The problem

Direct connections are faster, cheaper and more private than any detour — but nobody knows in advance where they hold up.

- **Every network is capable of something else**
  One Wi-Fi lets everything through, the next only port 443; on mobile the carrier-grade NAT decides. The same app is directly connected in one place and needs a detour in the next.
- **Browsers are not equal**
  Chromium, Firefox and WebKit differ in WebRTC details; on iOS whatever Safari allows is the whole story.
- **IPv6 saves the day — sometimes**
  Without a TURN server, peers behind restrictive NATs fail over IPv4, while the same connection over IPv6 is reliable.
- **Nobody measures this at scale**
  There are speed tests for bandwidth. There is no map for the question every P2P application has to ask: is direct enough here, or do I need a relay?

The insight

When two devices connect out-of-band via QR code — no relay, no signaling server — the connection attempt measures exactly what the network is capable of.

- **The instrument already exists**

  libp2p-webrtc-qr exchanges signed SDP payloads via QR code or link. The demo already reports which IP families both sides have.

- **One attempt, one data point**

  Success or failure, candidate types (host/srflx/relay), IPv4 vs IPv6, setup time, browser, network type, provider (ASN), coarse location.

- **The measurement costs nothing extra**

  It falls out of something the user wants to do anyway: connect two devices. No separate test, no extra app.

WEBRTC PUBLIC DATA MAP

How it works

Four steps, all on our own stack — from the measurement in the browser to the public map.

1 · Measure

- During the QR handshake, opt-in
- Signed locally with the user's WebAuthn DID
- No SSIDs, no MAC addresses, no GPS point

2 · Collect

- Measurements land in an OrbitDB
- The relay pins and archives (spin-off A)
- The user keeps a copy of their own data

3 · Aggregate

- Summary per network, cell and provider
- Three levels: direct works · mixed · relay needed
- Only above enough independent measurements

4 · Use

- Map public and free
- A concrete recommendation: direct is enough here, a relay there
- Raw data tradeable via UCAN delegation, revenue share to those who measured

Who pays for it

The map is free, and it is the shop window. What sells are the answers that follow from it — and the infrastructure it recommends.

Developers & platforms

- Video, gaming and P2P providers
- Want to know where direct is enough — and where relay capacity has to be budgeted
- API/SDK subscription, queried by provider and region

Network operators

- Hotels, municipalities, campuses, coworking, ISPs
- A report on their own network: what struggles, and why
- A re-measurement after the fix, evidencing "direct works here"

Research & regulation

- Net neutrality is unverifiable without measurements
- Universities, NGOs, regulators
- Dataset licence, open extracts for the public

> The real lever is in the first column: every red patch on the map is an evidenced recommendation for a relay — and therefore a sale for spin-off A. The map measures the demand our infrastructure covers.

Why us

The dataset is a by-product of technology we build and operate anyway.

- **We own the instrument**
  `@le-space/libp2p-webrtc-qr` is published, in beta and running as a demo — including signed payloads and a copy/paste fallback.
- **We already have the answer too**
  Where the map says "relay needed", the relay from spin-off A is one click away. No other measurement service can cover the demand it measures in the same breath.
- **Credibility on privacy**
  A data market only works if users trust the collector. We have spent years building systems that deliberately know little.

Status & next steps

The client exists, the map does not. The next steps are small and individually verifiable.

- **Client in beta** TODAY
  QR and link connections between browsers, signed SDPs, file transfer via Helia, live demo online.
- **Measurement schema & opt-in** IN PROGRESS
  Define which fields are collected, and a consent dialog you can understand in one sentence.
- **First map with a recommendation** IN PROGRESS
  Aggregation, a public map and the conclusion it produces per place: is direct enough, or budget a relay? Starting with our own tests and conferences.
- **Marketplace pilot** PLANNED
  One paying first customer, revenue share for users, anti-manipulation against forged measurements.

Open questions

Deliberately on their own slide: these four points decide whether the model holds.

- **Privacy**
  How coarse must the location be so "this network" never becomes "this flat"? GDPR assessment before the first measurement.
- **Legal side of rating**
  Public network ratings are lawful as long as they are evidenced and presented as a measurement rather than a verdict. A fee may only ever pay for a fresh measurement, never for an entry to disappear.
- **Manipulation**
  What stops a provider from measuring itself green? Signed measurements, reputation, outlier detection.
- **Critical mass**
  The dataset only sells above a certain density. Where do the first hundred thousand measurements come from?

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Let's talk

Wanted: partners with networks to measure — conferences, campuses, municipalities — and first customers who today have to guess whether a connection will succeed.

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Demo Source npm